

DailyConnect: A Mobile Aid that Assists the Understanding of Situation-based Emotions for Children with ASDs

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ABSTRACT

The understanding of situation-based emotions is of great significance for children, as it is able to assist children to better understand other people's behaviors in certain emotional scenarios and also to understand how their behaviors and actions can affect others. However, children with Autism Spectrum Disorders (ASDs) might have difficulties associating situations with emotions. In this paper, we presented the design of DailyConnect, a visual-based mobile application that supports children with ASDs to recall memories by reviewing photos based on their personal experience, and then taught by their teachers to understand situation-based emotions. There are two types of user modes for parents and teachers respectively in the system. For parents, the system enables them to upload daily-life photos associated with descriptions of contexts and behaviors of the child. For teachers, the system enables them to guide the children to recall what happened by reviewing the uploaded photos, and adopts the Discrete Trial Training (DTT) teaching technique to help the children understand situation-based emotions. Through recalling the memories of the children's real-life experience, the system enables teachers and parents to help children build emotions and situations associations step by step, including the understanding of contexts, emotional cues, emotions recognition, facial expression recognition and learn expected responses. Moreover, the system can provide individual feedback for each child to achieve targeted training after a long-term use: by analyzing of the data recording of the children (if with their consent), the system is able to generate reports on which step of the

understanding situation-based emotions needs attention for each child respectively.

KEYWORDS

Autism Spectrum Disorders, Understanding of situation-based emotions, Discrete Trial Training, Mobile application

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1 INTRODUCTION

Emotional understanding is the foundation of individual development and adaptation to society, as it plays an essential role in social communication and interaction. Situation-based understanding of emotions means that children are able to match emotions to specific situations. This skill can assist children to better understand why people behave the way they do in certain emotional scenarios and to understand how their behaviors and actions can affect others [1], not only a mechanical memorizing of emotions or facial expression recognition. For most children over 5 years old, they are already able to establish basic situations related to emotions [2]. However, children with ASDs lack such skills [3, 4]. ASDs are regarded as neurodevelopmental conditions characterized by impairments in associating emotions with situations and emotion recognition and expressions [5]. While recently ASDs are more defined as neurodiversity rather than neurodevelopment disorders [6] [7], improving their understanding of emotion situation would help them better adapt to the mainstream society.

1.1 Recent intervention approaches

Interventions may help children with ASDs to improve their understanding of emotion situation, especially by visual means [8]. Traditional interventions use paper-based tools such as cards or printed photos to support skills improvement [9, 10]. For example, a

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variety of training programs may ask children to label facial expressions from formatted photographs [11]. However, these approaches are resource-intensive [12] and are lack of recording and instant feedbacks to enable communications between parents and teachers.

Tablet-based or mobile-based tools have also been developed to support interventions of children with ASDs to improve their understanding of situation-based emotions. Compared with traditional tools, Tablet-based or mobile-based approaches are advantageous in the ways that (1) it is predictable (e.g. feedback), repetitive (e.g. rich resources), and portable which is favored by children and caregivers of children; (2) it supports instant-message record and review, which is supporting the communication between parents and teachers; (3) it offers an easy way to monitor children's learning performances and provide appropriate feedback, which can reduce the workload of teachers [7]. However, most tablet-based or mobile-based systems have focused on teaching children to recognize facial expressions or understand context from designed stories [13, 14]. For example, EmoStory [15] supports children's emotional development, including learning basic emotions and associated contexts and facial expressions through animated stories. However, children may still have difficulties in associating emotions with situations, as all these visual materials are "designed" instead of based on real-world scenarios they have experienced in real life.

1.2 Recent approaches based on memories recall

From the theories of memory, memory recall or retrieval is remembering the information or events that were previously encoded and stored in the brain [16]. Photos taken from the events of what the person has experienced could be context cues [17], which appear to be of great significance in the retrieval of newly learned meaningful information. Therefore, compared with "designed" and brand-new materials, to understand emotions based on visual materials from memories would cost less cognitive load to learn social skills and trigger more emotional empathy.

Therefore, we can assume that compared with existing approaches, to recall memories based on visual materials that the children have experienced, incorporated with the DTT [18], would be a more efficient approach to support children with ASDs to understand situation-based emotions.

A few recent tablet-based or mobile-based systems have adopted memories recall to support children with ASD to learn skills. For example, a picture-based approach [19] has been demonstrated that it is essential to improve the communication deficit among children with ASDs, while these approaches have been claimed that people use photos to trigger and reinforce their memories as well as to share their experiences in everyday life [20, 21].

However, these systems focused on language and speech training for children, or the communication between home and school, rather than to support children to understand situations and associate emotions with situations through memories recall by viewing photos taken from their daily life. Therefore, to overcome the drawbacks discussed above, we have identified the following core design philosophies for DailyConnect.

2 DAILY-CONNECT

We co-designed the DailyConnect application with our research team – which consisted of special education teachers and parents with children with ASDs. The research team conducted several discussions with teachers in a Special Education School in Anhui, China, the inquires-based discussions focused on understanding how to support children to understand situation-based emotions and how to support communication between the settings of home and school. The initial design came up after our interviews with the teachers and parents, and is refined based on their feedback over time. These interactive and iterative discussions lead to the design of the DailyConnect application to support collaboration between home and school, for the purpose to support children with ASDs to understand situation-based emotions.

2.1 Learning Goals

The main learning goal is to support children with ASDs (1) to understand the context and emotional cues; (2) to match emotions with situations (emotion recognition); (3) to match facial expressions with emotions (facial expression recognition); (4) to understand how his/her behaviors affect others. Since the system involves questions-and-answers based interactions, therefore the expected user group for the system should be the children with ASDs who have basic verbal communication skills.

2.2 Design Goals

We proposed these design goals for the system: **(1) Visual-based.** The system needs to support visual based content as photos uploaded from parents are designed to be presented as visual scenarios to help children to recall memories and understand contexts. **(2) Calendar-based.** The system should provide a calendar-based view; therefore, photos could be easily organized, reviewed and managed by dates. **(3) Collaboration between home and school.** The system is designed to enhance the collaboration between home and school, as parents are able to use the system to upload daily life photos, then teachers are able to review the photos with children, the system seamlessly enabled home-school collaboration with the help of modern Internet technology. **(4) Incorporated with the Discrete Trial Training (DTT).** DTT is a teaching technique that comes from Applied Behavior Analysis (ABA), which is found particularly effective for teaching skills to children with ASDs [18]. With the help of DTT, teachers can teach any behaviors or skills by breaking down the skill obtaining process into smaller steps, which makes it easier for the learners to grasp. In the light of DTT, the system breaks the process of understanding situation-based emotions into the following steps: 1. understanding the context; 2. understanding emotional cues; 3. understanding emotion recognition; 4. understanding facial expression recognition; 5. understanding response behaviors. **(5) Personalized tags.** The system should support personalized tags, allowing teachers to assign personalized tags to each child for personalized training, due to the different stimulus and symptoms of different children might have.

3 MAIN FUNCTIONS OF DAILY-CONNECT

There are two types of user modes in the system, for parents and teachers respectively, in a publisher-subscriber pattern. The parents

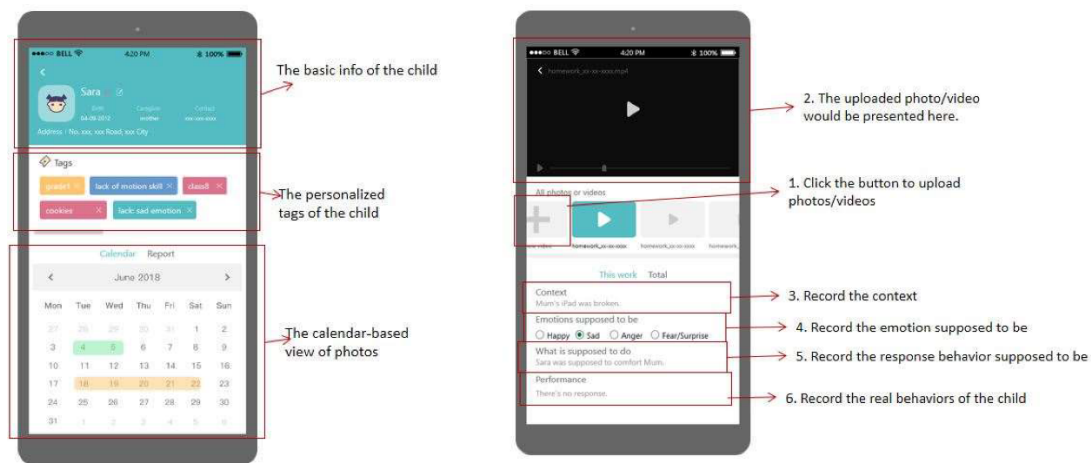


Figure 1: Examples of the profile page and the upload page (user mode: parents).

will act as a publisher - the system enables them to publish daily-life photos with associated information. The teachers will be the subscribers, as they can review photos published by parents, then adopt the DTT teaching technique and proceed with their teachings.

3.1 For parents

3.1.1 Profile. The profile page is shown as the left image on Figure 1. This page contains the basic information about the child (name, age etc.), the parents/care givers contact details, and personalized tags created by the teachers. In the bottom of the profile page, a personalized calendar is provided for parents to view uploaded photos.

3.1.2 Upload. In the upload page as shown in the right image on Figure 1, the parents can select a date on the calendar then select photos/videos for uploading, then filling in detailed information including context, emotions (e.g., happy, sad, anger, fear), expected response behaviors and actual response behaviors (All these steps are presented on the image). This information would be shared with the teachers through the system.

3.2 For teachers

3.2.1 Children List Page. The children list page is used for teachers to view the list of all children they teach, each list item with name and avatar. In this page, the teacher may access to the tags associated to each child. Different from the parents' mode, teachers have permissions to create and assign personalized tags for the children; the teachers also have access to each child's profile page and review photos uploaded by the parents. If the teacher approves the information submitted from the parents, he/she will be able to review the photo on the photo page with the child at school.

3.2.2 Photo review page. The photo review page is the key to the DTT teaching techniques employed by the teachers to teach a child to understand situation-based emotions. In the photo review page, the teacher would pick up a photo from the photo gallery (the left image on Figure 2) provided by the parents (The example photo on the Figure 2 was provided by a parent and allowed by the child.

Mosaic has been used to protect privacy), and guide the child to recall his/her personal experience on the moment when the photo was taken, and then go through each of the steps in DTT to teach the child to understand situation-based emotions. Afterwards, the teacher would fill in a form provided for the photo reviewed (see the examples as left and right image on Figure 2) in this page and record the child's performance during the teaching session.

Table 1 is an example of the form that reflect each of the steps in the DTT teaching techniques what the teacher and the child are supposed to go through during the photo review session in this system. The form is designed in a question-and-answer format. In Table 1, we assumed that all the responses the child provided are correct in all steps.

In each step, there would be several seconds for the child to respond. The step 1 is designed for the child to perform memories retrieval and provide guidance for the child to be aware of the context. If the child cannot provide an expected response, the teacher is supposed to intervene and tell the child what context is. The steps between step2 and step5 are about situation-based emotions and are reflected in the evaluation form in the page. If the child has the expected response, the teacher is supposed to click the tick button (the button would turn green) on the step (See the left image on the Figure 2) and give the child positive feedback, e.g., enthusiastic praise or a chosen reinforcer (e.g., a piece of cookies that the child likes).

If there is no response or the answer from the child does not match the situation ("incorrect"), the teacher is supposed to click the cross button (the button would turn red) on the step and provide a corrective feedback (See the right image on the Figure 2 and the Table 2). For example, the child could answer correctly about the context when viewing the photo, but no response to select the emotions (See the right image on the Figure 2), the teacher is supposed to point the sad facial expression with sentence "It would be a sad feeling, because mum's iPad was broken". During all these steps, responses from the child are recorded for teachers and parents to review and analysis.

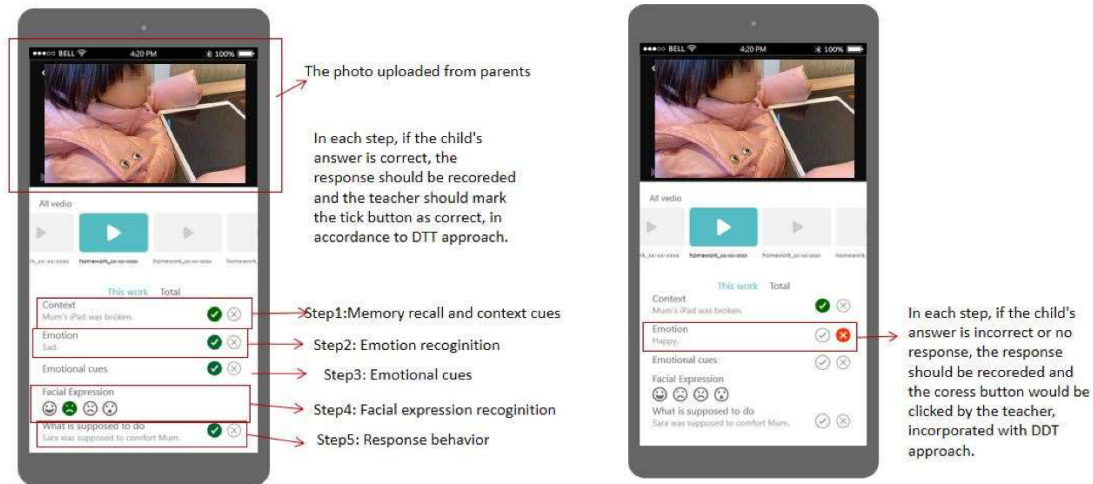


Figure 2: Examples of the photo review page incorporated with DTT (user mode: teacher).

Table 1: An example of the breakdown steps of DTT teaching techniques (in each step, we assume the response is correct)

The part supposed to be trained in this step	These examples are what the teacher is expected to act if child's responses are correct.
Step1: Memory recall and context cues	Teacher: Look. Could you please tell me where and what's this? Child: At home. The iPad is broken. Teacher: Yes. It's last Sunday. Your mum's iPad was broken. Can you remember? Child: Yes. I can.
Step2: Emotion recognition	Teacher: So, how's mum's feeling would be that time? Happy, sad, anger, surprise or fear? Child: Sad.
Step3: Emotional cues	Teacher: Yes. Mum felt sad that time. (Delivers reinforcer) Teacher: Would you mind telling me the reason why mum felt sad that time? Child: Coz her iPad was broken. She likes that iPad. Teacher: Good job!
Step4: Facial expression recognition	Teacher: OK. So, could you please select a facial expression to match sad? Child: (Pick one facial expression to match "Sad") Teacher: Yes. It's a sad face here! (Delivers reinforcer)
Step5: Response behaviors	Teacher: So, when mum's iPad is broken, what would you like to do? Child: I should comfort her. I can tell mum to fix it or to buy a new one. Teacher: Awesome! You are supposed to comfort mum if she feels sad. (Delivers reinforcer)

As the system is a mobile application, it supports repetitive teachings. Each teaching session can be replayed, therefore made it convenient for teachers and the child to repeat the session until the child could answer correctly in all these steps. If necessary and with consent, the voice from the teacher and the child during the DTT session can be recorded and stored in this system.

For each child after several sessions, the system can analyze his/her historical performance data, generate some extent of prediction and provide training suggestions targeted for each child. For both parents and teachers, they could click the "Report" button on the personal profile page to check the generated reports for the child (See the example as the left image on Figure 3).

For example, a child can answer most of the contexts correctly, but always shows no response to the emotions. The system would

notice it and suggest the teachers to focus the training more on the emotion recognition part. For another child, he/she might be able to answer most of the contexts and emotions correctly, but always fails on the facial expressions, the system would suggest the teacher to train him/her more on facial expression recognition (See the example as the right image on Figure 3). The teacher can also create and assign tags for these children (e.g., The tags of a child on the profile page on the left image of Figure 3) based on the data records and analysis for the purpose of personalized training. Therefore, the system can not only help children to understand situation-based emotions, but can also help the teachers to perform targeted training for each child respectively, after a period of use time.

Table 2: Another example of the steps of DTT teaching techniques (in each step, we assume no response or incorrect response)

No response	Incorrect
The teacher should guide the child to retrieve memories of the context	The teacher should guide the child to retrieve memories of the context
Teacher: So, how's mum's feeling would be that time? Happy, sad, anger, surprise or fear?	Teacher: So, how's mum's feeling would be that time? Happy, sad, anger, surprise or fear?
Child: (No response)	Child: Happy (Or other incorrect responses).
Teacher: It would be a sad feeling, because mum's iPad was broken. (Point the sad facial expression)	Teacher: No. It would be a sad feeling, because mum's iPad was broken. (Point the sad facial expression)
Teacher: Do you know the reason why mum felt sad that time?	Teacher: Do you know the reason why mum felt sad that time?
Child: (No response)	Child: She'd like to go shopping. (Incorrect answer)
Teacher: Look, because your mum's iPad was broken at that time.	Teacher: No. Because your mum's iPad was broken at that time.
Teacher: OK. So, could you please select a facial expression to match sad?	Teacher: OK. So, could you please select a facial expression to match sad?
Child: (No response)	Child: (Touch the "happy" face)
Teacher: (Point the sad facial expression) You need to touch the sad face.	Teacher: (Point the sad facial expression) No. Mum felt sad that time. This is a sad face.
Teacher: So, when mum's iPad is broken, what would you like to do?	Teacher: So, when mum's iPad is broken, what would you like to do?
Child: (No response).	Child: I need to cry (Or other incorrect responses)
Teacher: Well, if mum is sad, you are supposed to comfort her. You can tell her to fix it or buy a new one.	Teacher: No, honey. You need to comfort your mum. You can pat her on the shoulder, or tell her to fix it or buy a new one.

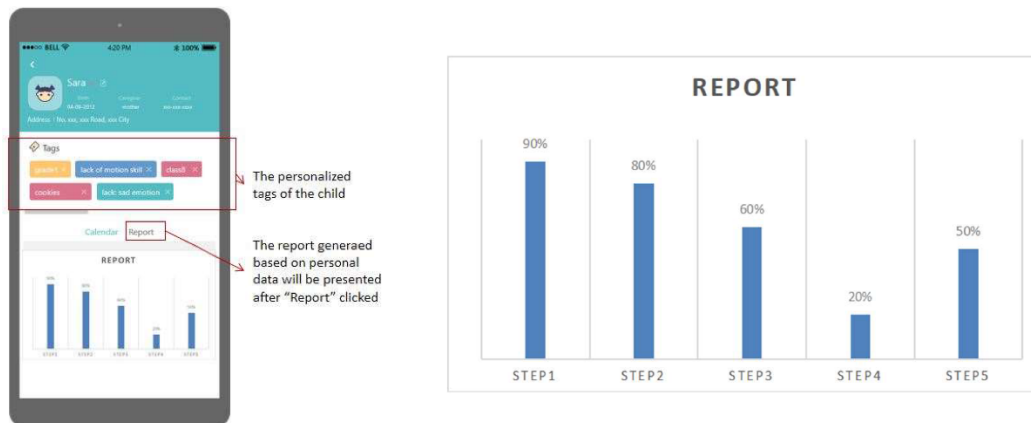


Figure 3: Examples of the reports generated based on personal data.

4 TECHNOLOGY SOLUTIONS

The system was designed on an Android platform. The prototype of the system was designed in React Native, which supports cross-platform deployment. The server was implemented using Java, and the data was stored on MySQL hosted on a cloud server, with security. We have considered to use speech recognition to analyze the responses from children, however, due to possible ambiguity in verbal speech and limited recognition patterns, we adopted the approach with teachers' manual intervention.

5 CONCLUSION

In this paper, we presented DailyConnect – a visual-based mobile application that supports children's understanding of situation-based emotions. Compared with previous systems or applications, our approach has these advantages: (1) The way that DailyConnect adopted to recall memories based on visual materials that children have experienced in daily life, would be a more efficient approach to support children with ASDs to understand situation-based emotions; (2) DailyConnect incorporated with the DTT teaching technique which was found particularly effective for teaching skills to children with ASDs; (3) The target audiences' usage data could be collected, reviewed and analyzed by teachers for the purpose of personalized and targeted training; (4) DailyConnect is designed

as a mobile application, which has the advantages of portable, predictable, repeatable and enhances collaborations between home and school.

6 FUTURE WORK

The next steps including completing implementation, conducting usability testing with children to refine the design, and then administering user studies to validate its use with children with ASDs. Specifically, we will conduct case studies to compared with traditional approaches, and find out how much DailyConnect can improve children's abilities in understanding of situation-based emotions. For the evaluation plan, we will measure the correctness in each of the steps during their usage of the system, then compare the variations of the correctness to measure the improvements on their understanding of situation-based emotions. In the long-term future, we will also collect and analyze the usage data, and summarize what kind of situation (e.g., relaxation situation which has high valence and low arousal emotions) may or may not affect the children's understanding of situation-based emotions.

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